# Daksh Arora

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## Education

- **Vellore Institute of Technology** — Chennai, India
  *B.Tech in Computer Science and Engineering; CGPA: 8.87* — *Sep 2024 - Jul 2028*
  ***Activities:*** *Technical Head, IEEE Computer Society VIT*

## Technical Skills

- **Languages**: Golang, TypeScript, JavaScript, Python, C, C++, Java
- **Frameworks**: Node.js, Express.js, Nest.js, FastAPI, Gin, React.js, TailwindCSS, OpenCV
- **Tools & Platforms**: Git, Docker, Postman, Firebase, Supabase, Linux, AWS (EC2, S3, SSM, Secrets Manager)
- **Databases**: PostgreSQL, MongoDB, MySQL, MariaDB, DynamoDB, Redis, Firebase Firestore
- **Concepts**: REST APIs & Webhooks, WebSockets, Event-Driven Architecture, Outbox Pattern, CI/CD, pytest

## Experience

- **Mintzy** — Remote
  *Quant Software Developer Engineer Intern* — *Oct 2026 - Present*
  - **Trading-signal service**: Built an unattended, event-driven algorithmic trading-signal service in Python (roughly 11,000 lines) that converts AI price predictions into live long/short signals for NSE equities and routes them to the Tradetron and Algoninja brokers every trading day — now live with real broker delivery across 8 stocks on an INR 10 lakh notional book.
  - **Paper-trading engine**: Simulated fills, slippage, a cash ledger, realized/unrealized P&L, position flips and end-of-day square-off on 75-minute candles, emitting each position change as a sequenced structured signal event.
  - **Reliable delivery layer**: Engineered a durable MongoDB outbox with per-destination dispatcher threads, backoff retries, rate limiting and a time-based send gate, plus a reconciler verifying each broker's state against the engine's book.
  - **Market data & testing**: Integrated Angel One SmartAPI real-time prices over WebSocket (unattended TOTP login with stale-tick guards) and ported a quantitative capital-allocation strategy from research into production; shipped 282 pytest tests (unit, property-based, end-to-end) and a FastAPI health/status API.
- **Independ** — Remote
  *Software Development Engineer Intern* — *June 2026 - Present*
  - **Backend testing**: Wrote 1,700+ lines of Vitest unit tests across the files and admin modules (schedulers, services, repositories) in a NestJS backend, validating CASL-based access control, cursor pagination and dependency injection.
  - **AI guardrails**: Built guardrails for the AI pipeline — JSON-schema-constrained structured outputs, strict Zod validation and retry-on-invalid-output logic — plus an LLM negotiations-context module enriching prompts with authenticated history and structured bid data.
- **TX Minds** — Remote
  *Software Developer Intern* — *Apr 2026 - Sep 2026*
  - **Payments backend**: Built the donation and payment backend for a non-profit platform (Node.js, TypeScript, Express 5, Drizzle ORM, PostgreSQL), integrating HDFC and ICICI payment gateways, 2FA, Cloudflare Turnstile and a custom Strapi CMS OTP-auth plugin.
  - **Async receipt pipeline**: Designed an idempotent, rate-limited BullMQ and Redis job pipeline with exponential-backoff retries, delivering donation receipts and OTP/password-reset emails via AWS SES.
  - **Cloud migration & CI/CD**: Led the migration from Render to AWS (EC2 behind an ALB with ACM TLS, RDS PostgreSQL, S3) and built a GitHub Actions pipeline deploying via ECR, OIDC and SSM with no stored keys, health-check gating and automatic rollback.

## Projects

- **Peer-to-Peer File Sharing System (Golang, TCP Sockets, Distributed Systems)**: Built a distributed P2P file-sharing system with deterministic chunking, content-addressable storage and hash-based integrity verification, plus a custom TCP layer with length-prefixed framing and versioned handshakes for fault-tolerant, backward-compatible transfer.
- **Lore — On-Device AI Knowledge Assistant (Python, FastAPI, RAG, LanceDB, Snapdragon NPU)**: Built an on-device RAG assistant (document parsing, semantic chunking, vector search) during the Qualcomm Snapdragon Multiverse Hackathon, optimizing inference on Snapdragon NPUs via Qualcomm AI Hub and eliminating cloud dependency while improving latency.

## Honors and Awards

- Winner, Qualcomm Snapdragon Multiverse Hackathon — built Lore, an on-device AI assistant.
- Top 30, Reverse Coding X, a reverse-coding competition hosted by IIT Madras as part of Shaastra.
- Technical Head, IEEE Computer Society VIT — hackathons, coding contests and workshops.